

Class 5: Data Viz with ggplot

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Table of contents

Background	1
Gene Expression Plot	4
going further with gapmider	6
First look at the dplyr package	8

Background

there are lot's of ways to make plots in R i.e lots of graphics in R. These include so called “base R” (like the `plot()`) and add on packages like **ggplot2**

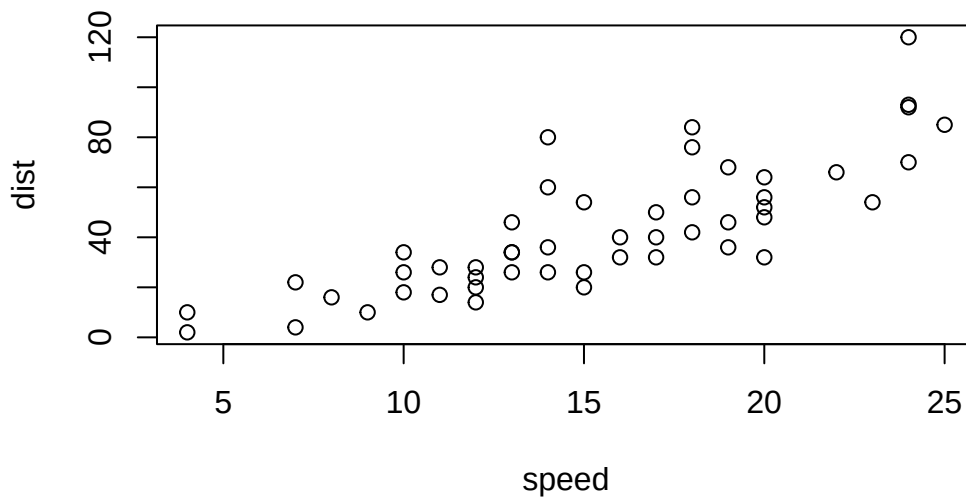
Let's make the same plot with these two graphics system. We can use the inbuilt `cars` dataset:

```
head( cars )
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

With “base R” we can simply:

```
plot(cars)
```

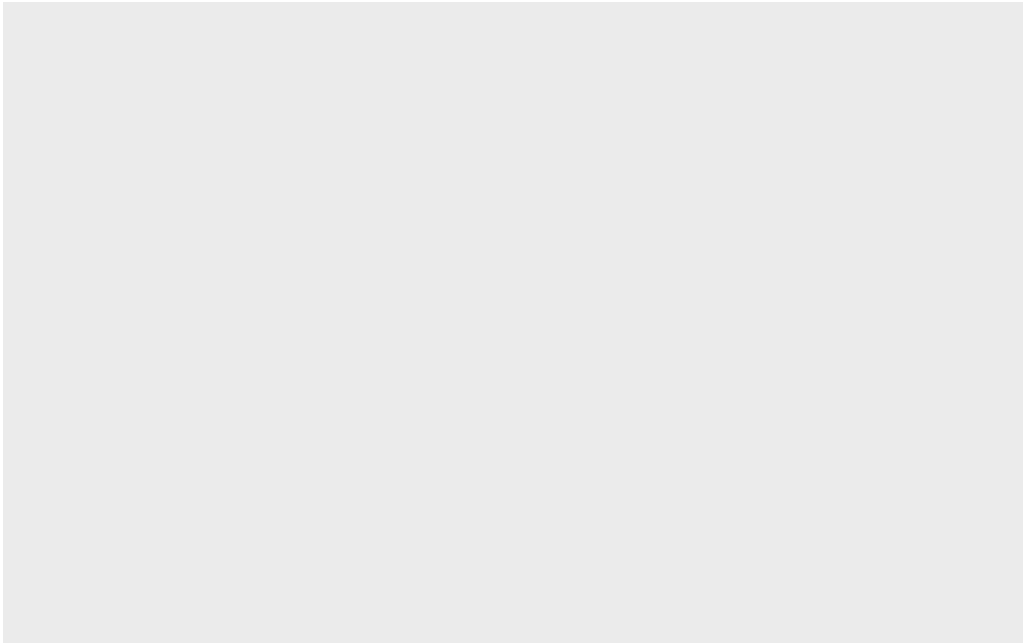


Now let's try ggplot. First I need to install the package using `install.packages("ggplot2")`

N.B. we never run an `install.packages()` in a code chunk otherwise we will re-install needlessly everytime we render our document.

every time we want to use an add-on package we need to load it up with a call to `library()`

```
library(ggplot2)
ggplot(cars)
```

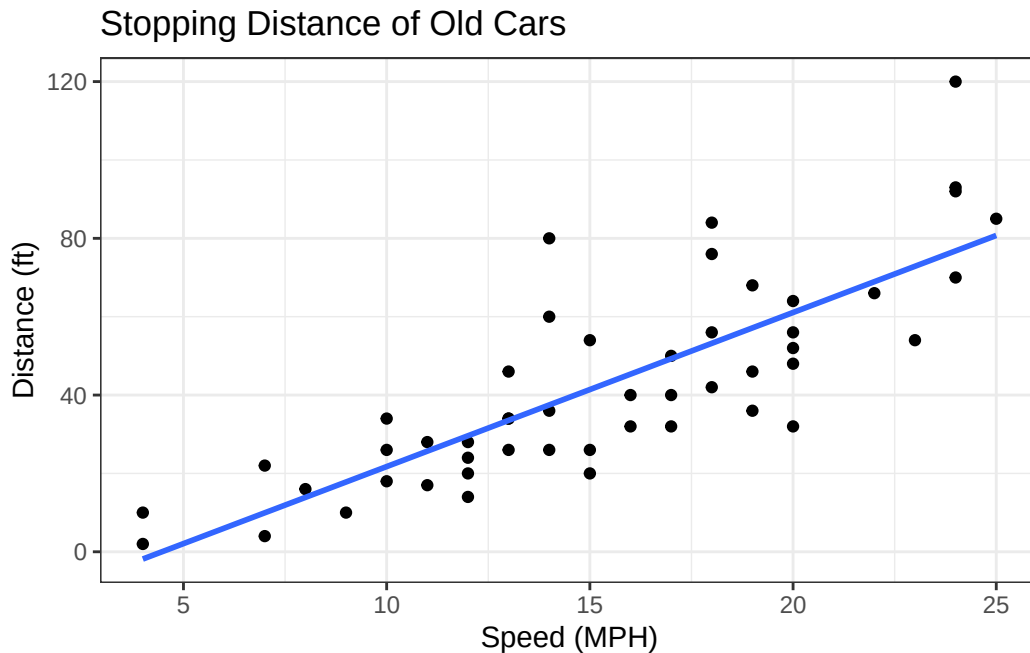


every ggplot needs at least 3 things:

1. The **data** i.e. stuff to plot as a data.frame
2. The **aes** or aesthetics that map the data to the plot
3. The **geom__** or geometry i.e the plot type such as points, lines etc.

```
ggplot(cars) +  
  aes(x=speed, y=dist)+  
  geom_point() +  
  geom_smooth(method = "lm", se = FALSE) +  
  labs(x="Speed (MPH)",  
       y="Distance (ft)",  
       title="Stopping Distance of Old Cars")+  
  theme_bw()
```

`geom\_smooth()` using formula = 'y ~ x'



Gene Expression Plot

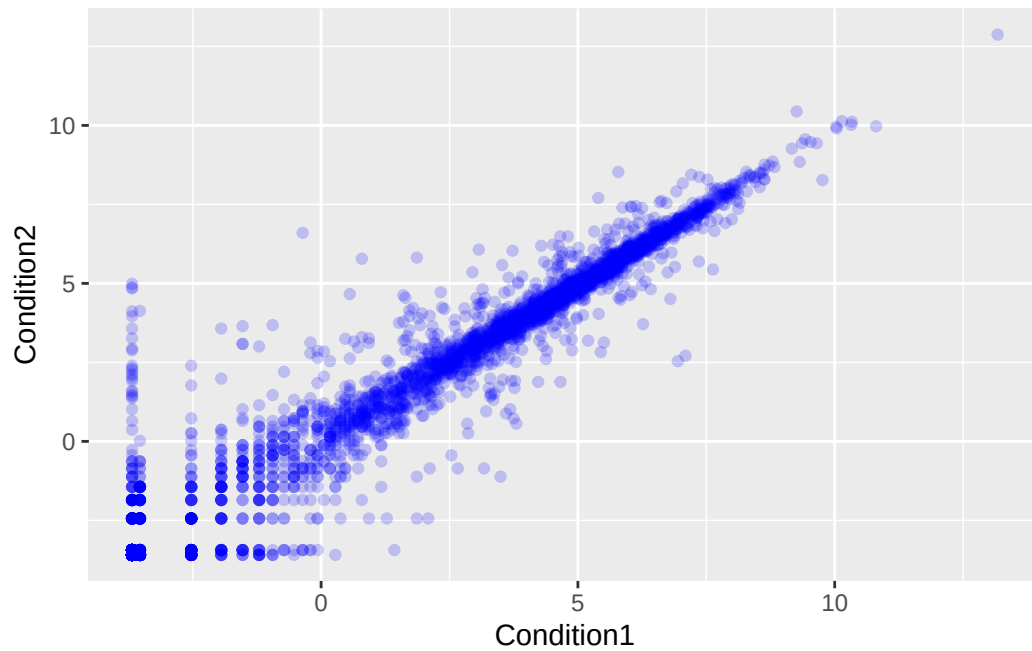
read some data on the effects of GLP-1 inhibitor (drug) on gene expression values:

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

	Gene	Condition1	Condition2	State
1	A4GNT	-3.6808610	-3.4401355	unchanging
2	AAAS	4.5479580	4.3864126	unchanging
3	AASDH	3.7190695	3.4787276	unchanging
4	AATF	5.0784720	5.0151916	unchanging
5	AATK	0.4711421	0.5598642	unchanging
6	AB015752.4	-3.6808610	-3.5921390	unchanging

Version 1 plot - start simple by getting some ink on the page.

```
ggplot(genes)+
  aes(Condition1, Condition2) +
  geom_point(col="blue",alpha=0.2)
```



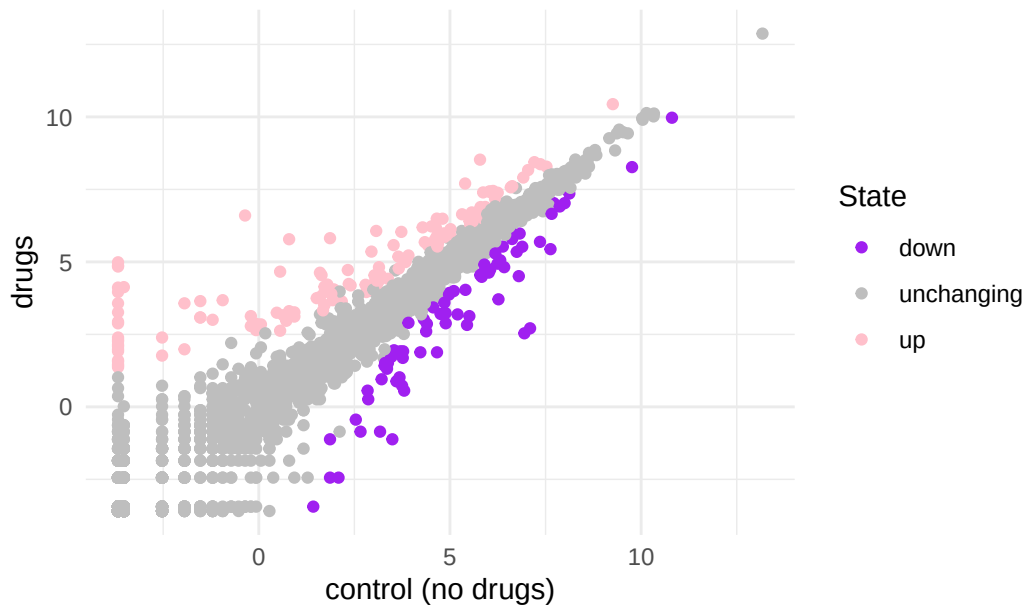
Let's color by State up, down or no change

```
table(genes$State)
```

down	unchanging	up
72	4997	127

```
ggplot(genes)+
  aes(Condition1, Condition2, col=State) +
  geom_point()+
  scale_colour_manual( values=c("purple","gray","pink") )+
  labs(x="control (no drugs)",
       y="drugs",
       title="expression changes upon administration of GLP-1 drug")+
  theme_minimal()
```

expression changes upon administration of GLP-1 drug



going further with gapminder

here we explore the famous `gapminder` dataset with some custom plots

```
# File location online
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv"

gapminder <- read.delim(url)
head(gapminder)
```

	country	continent	year	lifeExp	pop	gdpPercap
1	Afghanistan	Asia	1952	28.801	8425333	779.4453
2	Afghanistan	Asia	1957	30.332	9240934	820.8530
3	Afghanistan	Asia	1962	31.997	10267083	853.1007
4	Afghanistan	Asia	1967	34.020	11537966	836.1971
5	Afghanistan	Asia	1972	36.088	13079460	739.9811
6	Afghanistan	Asia	1977	38.438	14880372	786.1134

Q. how many rows does this dataset have?

```
nrow(gapminder)
```

```
[1] 1704
```

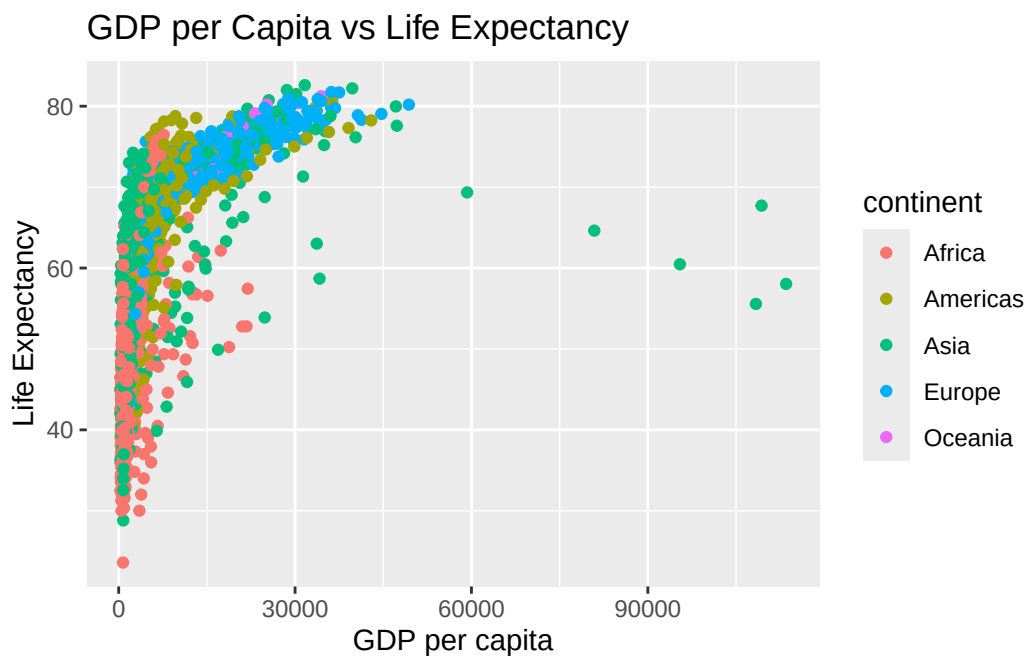
How many different continents are in this dataset?

```
table(gapminder$continent)
```

Africa	Americas	Asia	Europe	Oceania
624	300	396	360	24

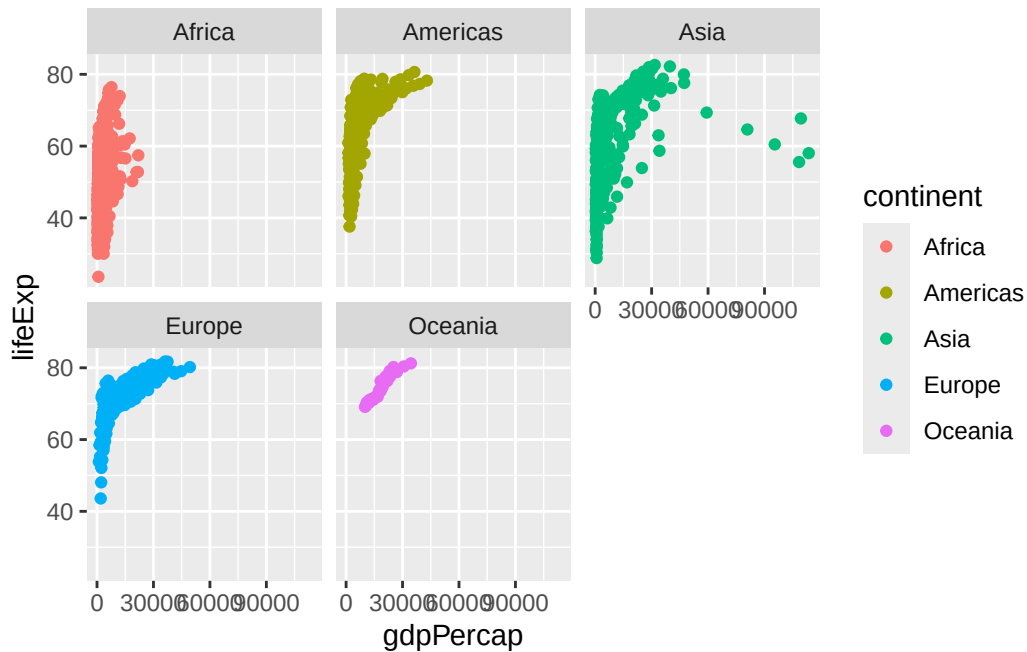
Version 1 plot gdpPerCap vs LifeExp for all rows

```
ggplot(gapminder)+  
  aes(gdpPerCap, lifeExp, col=continent)+  
  geom_point()+  
  labs(x="GDP per capita",  
       y="Life Expectancy",  
       title = "GDP per Capita vs Life Expectancy")
```



I want to see a plot for each continent - in ggplot lingo this is called “faceting”

```
ggplot(gapminder)+
  aes(gdpPercap, lifeExp, col=continent)+
  geom_point()+
  facet_wrap(~continent)
```



```
labs(x="GDP per capita",
     y="Life Expectancy",
     title = "GDP per Capita vs Life Expectancy")
```

```
<ggplot2::labels> List of 3
 $ x      : chr "GDP per capita"
 $ y      : chr "Life Expectancy"
 $ title  : chr "GDP per Capita vs Life Expectancy"
```

First look at the dplyr package

another add-on package with a function called `filter()` that we want to use.

```
library(dplyr)
```


Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

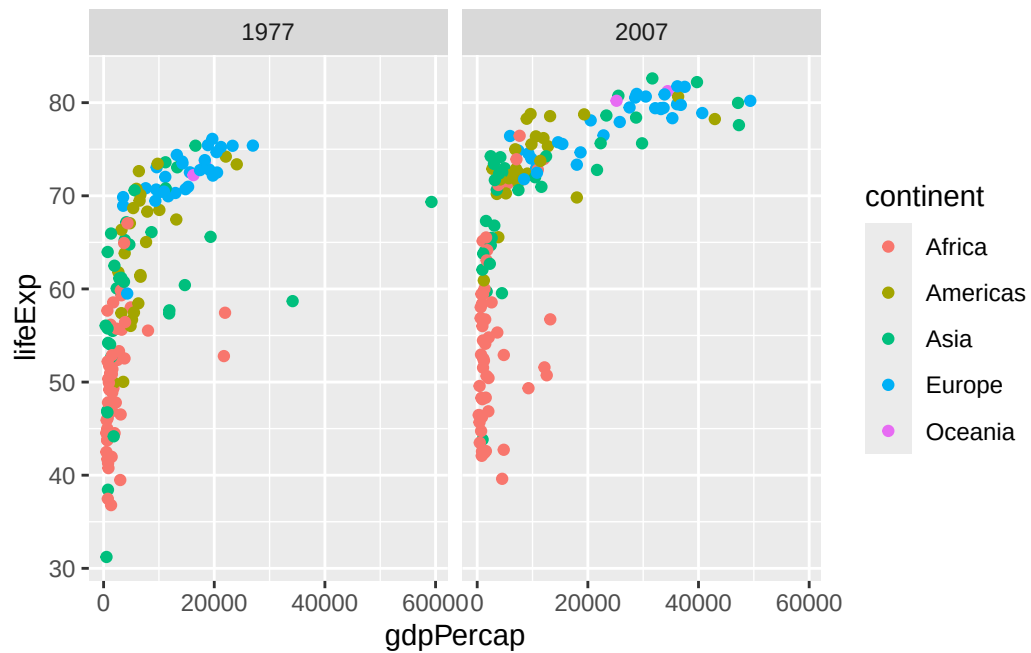
```
filter(gapminder, year==2007, country == "United States")
```

	country	continent	year	lifeExp	pop	gdpPercap
1	United States	Americas	2007	78.242	301139947	42951.65

```
input <- filter(gapminder, year == 2007 | year == 1977)
```

“|” means “or”

```
input <- filter(gapminder, year == 2007 | year == 1977)
ggplot(input)+
  aes(gdpPercap, lifeExp, col=continent)+
  geom_point()+
  facet_wrap(~year)
```



```
labs(x="GDP per capita",
      y="Life Expectancy",
      title = "GDP per Capita vs Life Expectancy")
```

```
<ggplot2::labels> List of 3
 $ x      : chr "GDP per capita"
 $ y      : chr "Life Expectancy"
 $ title  : chr "GDP per Capita vs Life Expectancy"
```